THE FLAT AT A POINT FUNCTOR IS UNCHANGED BY A FLAT LOCAL EXTENSION OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The situation defining the flat at a point functor. Same chapter flat.tex, situation labelled situation-flat-at-point (statement, proof lines 6109-6110 and 6118-6121). Fixes the category and the notation for the base changes of the algebras and of the module.
- The description of the closed set where two extended maximal ideals generate the unit ideal, and surjectivity of the induced map on those sets. Same chapter flat.tex, the displayed identifications at proof lines 6122-6127 together with the faithful flatness remark at proof lines 6128-6136. Identifies the two closed sets with spectra of tensor products of residue fields and shows that the map between them is surjective because the ring map between those tensor products is faithfully flat.
- Descent of flatness along a flat local homomorphism. Algebra, label algebra-lemma-flatness-descends-more-general, invoked at proof line 6141. Shows that for a prime of the extended target lying over a prime of the extended source, flatness of the localised source module over the test algebra is equivalent to flatness of the localised target module.

Source reference. Original lemma 0584: sources/stacks-chow/tags.txt, source lines 6767–6767. Navigation pointer for \ref{algebra-lemma-flatness-descends-more-general} used inside the counted span (line 6141); the argument that consumes it is retained in the span.

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Source reference. Original reference 05MK: sources/stacks-flat/pinned-original-source.tex, source lines 6008–6008. Navigation pointer for \ref{situation-flat-at-point} used inside the counted span (line 6110); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 05MK suppose that $B \to C$ is a flat local homomorphism of local rings. Set $N = M \otimes_B C$. Denote $F'_{lf} : \mathcal{C} \to Sets$ the functor associated to the pair (C, N). Then $F_{lf} = F'_{lf}$.*

Proof. Let A' be an object of $\mathcal{C}$. Set $C' = C \otimes_A A'$ and $N' = N \otimes_A A' = M' \otimes_{B'} C'$ similarly to the definitions of B', M' in Situation 05MK. Note that

$$V(\mathfrak{m}_{A'}B' + \mathfrak{m}_B B') = \mathrm{Spec}(\kappa(\mathfrak{m}_B) \otimes_A \kappa(\mathfrak{m}_{A'}))$$

and similarly for $V(\mathfrak{m}_{A'}C' + \mathfrak{m}_C C')$. The ring map

$$\kappa(\mathfrak{m}_B) \otimes_A \kappa(\mathfrak{m}_{A'}) \longrightarrow \kappa(\mathfrak{m}_C) \otimes_A \kappa(\mathfrak{m}_{A'})$$

is faithfully flat, hence $V(\mathfrak{m}_{A'}C'+\mathfrak{m}_C C') \to V(\mathfrak{m}_{A'}B'+\mathfrak{m}_B B')$ is surjective. Finally, if $\mathfrak{r} \in V(\mathfrak{m}_{A'}C' + \mathfrak{m}_C C')$ maps to $\mathfrak{q} \in V(\mathfrak{m}_{A'}B' + \mathfrak{m}_B B')$, then $M'_{\mathfrak{q}}$ is flat over A' if and only if $N'_{\mathfrak{r}}$ is flat over A' because $B' \to C'$ is flat, see Algebra, Lemma 0584. The lemma follows formally from these remarks. □